



Spring Research Project: Mentor Rubric

Mentor Name(s): Melanie Zaidel

Mentee Name(s): Vivian Repka

Project Title: Outnumbered by the Invisible: Calculating the amount of Galactic Dark Matter

In a short paragraph, explain the project and its learning objectives:

In this project, Vivian will learn about the basics of dark matter and focus on the calculation of how much dark matter is in the Milky Way Galaxy. First, she will watch videos about the basics of dark matter and discuss details with me. Next, we'll discuss dark matter density profiles, and create a plot of a cored-NFW profile. After that, we will discuss numerical integration in Python to evaluate the integral of the density profile and determine the mass of dark matter in the Galaxy. Finally, I will guide Vivian through a derivation in celestial geometry to parameterize line-of-sight integrals for calculating the column density of dark matter in different directions. Time permitting, she may plot these column densities on a skymap.

What are the deliverables? (i.e. plot/figure/image/equation mentee will reach)

1. Plot of dark matter density as a function of radius for a cored-NFW profile.
2. The amount of dark matter in the Galaxy inferred by the NFW profile, given in solar masses.
3. An equation describing the dark matter column density as a function of direction in terms of a line-of-sight integral.
4. Time permitting: A skymap showing the dark matter column density as a function of Galactic coordinates.



Project Calendar: Indicate what the mentee will be expected to do each week. Work days are indicated in the left column. Include estimated time, resources, in/out of class work, and whether it is completed on their own or with you. The project should take 8-10 hours in total.

Dates:	Tasks:
Week 1 (2/16) Intro to project	1 hr on own: Watch videos about dark matter
Week 2 (2/23) In class work day	In class: Mentor family meeting to discuss what you've learned with Finn 1 hr with mentor: Discussion about videos, introduce NFW profile and begin plotting workbook
Week 3 (3/2) In class work day	1 hr on own: Finish NFW plot and begin numerical integration workbook
Week 4 (3/9) Outside of class work	1 hr with mentor: Finish numerical integration workbook and discuss total Galactic dark matter mass with mentor
Spring Break	None
Week 5 (3/23) In class work day	1 hr with mentor: Begin celestial geometry derivation
Week 6 (3/30) Outside of class work	1 hr with mentor: Continue celestial geometry derivation
Week 7 (4/6) In class work day	2 hr with mentor: Finish celestial geometry derivation, start working on poster
Week 8 (4/13) Poster work day	2 hr with mentor: Finish working on poster and rehearse presentation
Week 9 (4/20) Poster presentation!	

Reading List: (Make sure material is level appropriate.)

- YouTube videos on dark matter
- Melanie's basic Python and plotting tutorials



- Melanie's bachelor's thesis

Resource List: (This should include lab space, instruments, and software. Note that mentees are not allowed to complete research projects that require safety training operation of lab equipment. If you feel this is detrimental to your project, please petition Polaris leadership.)

- Personal computer with an installation of Anaconda Navigator

Other Notes:

Hello Polaris Leadership family I love you